MILITARY SPECIFICATION

CASTINGS; PEARLITIC MALLEABLE IRON FOR ARTILLERY AND MORTAR AMMUNITION

1. SCOPE

1.1 Scope. This specification covers pearlitic malleable iron castings for artillery and mortar ammunition components.

1.2 Classification. The castings shall be manufactured from the following grades of pearlitic malleable iron: 80002, 70002, 60003, 53004, 50007, 48004, 45010, 45007.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–A–2550 —Ammunition and Special Weapons, General Specification for

MIL–P–10025 —Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification for

MIL–I–45208 —Inspection Requirements, General Specification for

STANDARDS

FEDERAL

Federal Test Method Std. No. 151

MILITARY

MIL–STD–109 —Quality Assurance Terms and Definitions

Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Melting process. The iron shall be made by one of the following processes: air-furnace, open hearth, electric-furnace or any of the multiplex processes.

3.2 Molding. Molds and cores shall be made on semiautomatic or automatic molding equipment, approved by the procuring activity to assure that molds of consistent properties will be obtained on a reproducible basis.

3.3 Heat-treatment. Castings shall be annealed in a controlled atmosphere furnace which has been approved by the procuring activity. The specified properties shall be obtained by liquid quenching from an appropriate temperature followed by tempering. Heat-treatment shall be performed on the whole casting in such a manner as to result in uniform mechanical properties throughout the individual part and throughout the heat-treat lot.

3.3.1 *After heat-treatment.* After heat-treatment, heating of the castings above the tempering temperature shall not be permitted.

3.4 Mechanical properties. The castings shall meet the mechanical property requirements shown in table I.

3.5 Chemical composition. The chemical

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TABLE I. *Mechanical property requirements*

Grade	Yield strength (psi) minimum (0.1 % offset)	Tensile strength minimum (psi)	Elongation minimum (%)	Brinell* hardness
80002	80,000	100,000	2	241–269
70002	70,000	90,000	2	228–269
60003	60,000	80,000	3	197–255
53004	53,000	80,000	4	197–241
50007	50,000	75,000	7	179–228
48004	48,000	70,000	4	163–228
45007	45,000	68,000	7	163–217
45010	45,000	65,000	10	163–207

* Brinell hardness range is advisory only.

composition shall be such that the finished product possesses the required properties throughout and shall be that of an iron-base alloy containing carbon and silicon as the principal alloying elements.

3.6 **Foundry control.** Castings shall be produced under foundry control approved by the procuring activity. Foundry control shall consist of examination of castings by radiographic, micrographic, macrographic, ultrasonic or other approved means, until the pattern and mold design, and melting, molding and pouring practices have been established to produce castings meeting the quality standards furnished or approved by the procuring activity. Foundry control shall also consist of those process controls necessary to assure that castings of the required quality will be consistently obtained. When foundry practice and process controls have been so established, production practices shall not be changed without demonstrating to the satisfaction of the procuring activity that the change does not adversely affect the quality of the casting.

3.7 **Soundness.** The soundness of castings shall meet standards furnished or approved by the procuring activity using inspection methods specified in the item specification and by procuring activity. The quality standard shall consist of a sample casting or castings, radiographs, photomicrographs, macrographs, other standards specified by the procuring activity or a combination of these which shall clearly indicate the quality to be obtained.

3.8 **Repair of castings.** Defective castings shall not be plugged or welded.

3.9 **Surface finish and condition**

3.9.1 *Machined surfaces.* Machined surfaces shall meet the surface finish requirements specified on the applicable drawings.

3.9.2 *As-cast-surfaces.* As cast surfaces on the finished part shall be as specified on the applicable drawing and specification for the end item and shall be free of scale resulting from the casting process and subsequent heat-treatment operations. Surface defects such as pits, craters, sand holes, porosity, core line impressions or any similar defects than those permitted by the standards which are provided or stipulated by the procuring activity shall be cause for rejection of the casting. Grinding to remove surface defects shall be permitted to minimum thickness requirements provided that thickness measurements are taken in each ground area to insure compliance with drawing requirements. Grinding shall be done in such a manner as to provide a smooth contour and shall have no sharp corners or edges.

3.10 **Decarburization.** Unless otherwise specified by the procuring activity, the depth of decarburization and affected microstructure measured from any cast surface shall not be greater than .035 inches, when tested in accordance with paragraph 4.3.4.

3.11 **Mottle.** Castings shall be free of mottle (primary graphite) when tested in accordance with 4.3.3.

3.12 **Dimensional tolerances.** Dimensions and dimensional tolerances of finished castings shall be as specified in the applicable drawing.

3.13 **Identification markings.** The identification mark of the manufacturer, pattern number, date and time of casting shall be cast into the surface of each casting using a format and location as approved by the procuring activity.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to MIL–STD–109 to define quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 Inspection provisions.

4.2.1 *Lot.* A lot shall consist of castings of the same pattern, grade and melt or if cast by a continuous melting process, the lot may consist of castings that can be identified as having been poured during a designated portion of a working shift.

4.2.1.1 *Heat-treatment lot.* To determine compliance with the mechanical properties 3.4, any quantity of castings not to exceed four (4) hours production of the final heat-treatment in the same furnace, for the same conveyor speed, shall be considered as a heat-treatment lot. The lot thus constituted shall be the basis for selection of sample castings to be used for the tension tests, 4.3.1.2.

4.3 Inspection testing.

4.3.1 *Mechanical properties.*

4.3.1.1 *Hardness.* In order to insure uniformity of heat-treatment and to establish a hardness range, to assure compliance with specified mechanical properties each casting after final heat-treatment shall be subjected to a hardness test with a Brinell Hardness Tester at a location on the casting specified or approved by the procuring agency.

4.3.1.1.1 *Alternate hardness testing.* A direct reading Brinell Hardness Testing Machine may be substituted for the Brinelloscope method, provided the same diameter ball is used and the same load is applied as is used in Standard Brinell testing, and provided that at least 2000 castings are tested by both methods to establish a correlation between the two methods. After a correlation has been established, the diameter of the indentation in every tenth casting shall be measured with the Brinelloscope to assure continued correlation.

4.3.1.2 *Tension test.* Two castings from each lot representing the hardest and softest castings as determined by the hardness test outlined in 4.3.1.1 or 4.3.1.1.1 shall be subjected to the tension test. Two specimens from each casting shall be prepared as specified in 4.4.1.2 and tested. If all specimens comply with the requirements of 3.4, the hardness range for the lot will have been established. Failure of one or more tension test specimens to comply with the requirements shall be cause for rejection of the lot represented by the specimen.

4.3.1.3 *Retest.* When one or more tension test specimens fail to comply with the requirements of 3.4 and do not qualify for retest under 4.4.1.2.3, a new hardness range shall be established. The rejected lot may at the direction of the contractor, be reheat-treated prior to establishing a new hardness range. The tension test, in compliance with 4.3.1.2 shall be conducted to determine the acceptability of the new hardness range. When the hardness range has been established, all castings whose values fall outside this range shall be rejected.

4.3.2 *Soundness.* Each projectile body casting or any other casting that is subject to being loaded with high explosives or chemical filler shall be tested in accordance with 4.4.2 or as prescribed by the procuring activity. Failure of any casting to comply with 3.7 shall be cause for the rejection of the casting. Inspection and sampling on other ammunition components shall be as specified by the procuring activity and failure to com-

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ply with 3.7 shall be cause for rejection of the individual casting. When inspection is performed on a sampling basis the lot shall be rejected if the rejection criteria has been reached or exceeded.

4.3.3 *Mottle.* A minimum of two castings selected for every thirty minutes or fraction thereof of continuous pouring time shall be tested in accordance with 4.4.3. Failure of any sample to comply with 3.11 shall be cause for rejection of all castings poured since the last acceptable samples were taken.

4.3.4 *Decarburization.* One casting from each heat-treatment lot shall be tested in accordance with 4.4.4. Failure of the sample to comply with 3.10 shall be cause for rejection of the lot. Lots of castings that have complied with the requirements of 3.10 and are subsequently reheat-treated shall be reexamined as prescribed above.

4.3.5 *Identification markings.* At least one casting from each pour shall be visually examined for legibility of markings and for proper location to determine compliance with 3.13.

4.4 Test methods and procedures.

4.4.1 *Mechanical properties.*

4.4.1.1 *Hardness.* The Brinell hardness test shall be performed in accordance with procedures prescribed in Federal Test Method Std. No. 151. Due to the nature of the microstructure of pearlitic malleable iron, indenter balls of less than ten millimeter diameter shall not be used.

4.4.1.2 *Tension test.* From each sample selected in accordance with 4.3.1.2 two tension test specimens shall be taken longitudinally from each side of the sample, 180 degrees apart or as prescribed by the procuring activity.

4.4.1.2.1 *Tension test specimen.* The specimen shall be the largest obtainable standard round specimen prescribed by Federal Test Method Std. No. 151 (types R1 thru R5) which the size of the casting and location of the specimen will permit. If the size or geometry of the casting does not permit a standard round specimen, a specimen as specified or approved by the procuring activity shall be used.

4.4.1.2.2 *Yield strength, tensile strength, and elognation.* Yield strength shall be determined by the offset method as defined in Federal Test Method Std. No. 151. The offset shall be 0.1 percent (0.001 inches per inch of gage length). Tensile strength and elongation on each specimen shall be determined by utilizing the criteria for these tests prescribed by Federal Test Method Std. No. 151.

4.4.1.2.3 *Replacement test specimens.* If a tension test specimen fails to meet the requirements of 3.4 due to an identifiable defect in the tensile specimen which is acceptable under provisions of 3.7 and 4.3.2 or due to the fracture occurring under those conditions allowing for retests for elongation in Federal Test Method Std. No. 151, another specimen shall be machined from the same casting and tested. In the instance of failure due to an acceptable metal defect only one replacement specimen shall be permitted and failure of the replacement specimen to comply with the requirements of 3.4 shall be cause for rejection of the represented lot. Results of all tests, original and replacement tests, shall be recorded and specimens which failed to meet the requirements due to the above conditions shall be identified and retained by the contractor and made available to the procuring activity for examination upon request.

4.4.2 *Soundness.* For those applications where 100% inspection is required, castings will be inspected by ultrasonics or, if the geometry of the casting does not permit the use of ultrasonics, some form of radiography and Magnaglo, if approved by the procuring activity. Inspection methods for less critical components shall be as specified by the procuring activity.

4.4.3 *Mottle.* The samples selected in accordance with 4.3.3, shall be broken in such a manner that the fracture will extend through the heaviest section. The fracture shall be visually examined for compliance with 3.11.

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4.4.4 *Decarburization.* From the samples selected in accordance with 4.3.4, a suitable specimen shall be taken and prepared for study of the microstructure. The depth of decarburization shall be measured by viewing the specimen at 100 magnifications (100X). The zone of affected microstructure to be examined, to determine compliance with 3.10, shall include any structure which is essentially different from the matrix structure of pearlitic malleable iron such as ferritic or pearlitic rims and areas of lower nodule count or smaller nodules of temper carbon.

5. PREPARATION FOR SHIPMENT

5.1 Packing

5.1.1 *Level A.* Not applicable.

5.1.2 *Level B.* Not applicable.

5.1.3 *Level C.* Castings shall be packed in accordance with the requirements of MIL–P–10025.

5.2 Labeling and marking. Labeling and marking shall be in accordance with the requirements of MIL–P–10025.

6. NOTES

6.1 Intended use. Pearlitic Malleable Iron castings purchased to this specification are intended for use in Artillery and Mortar Ammunition.

6.2 Tension test specimens. Separately cast test coupons for tension test specimens are not always representative of the mechanical properties present in the casting and therefore shall not be used in tension tests for determining compliance with mechanical property requirements unless specifically authorized by the procuring activity.

6.3 Machining of test specimens. Due to the lower ductility of this material in the higher strength grades small test specimens, i.e., R4 and R5 types Federal Test Method Standard 151, may not represent the true elongation of the material if a reduced section of 0.1 percent of the dia. is used in the gage length. For this reason it is advisable to machine specimens without the permissible reduced section in the center of the gage length.

6.4 Government owned tooling. Due to the technical aspects involved, the mold patterns and core boxes shall be Government owned tooling.

6.5 Data cards. When specified by the contracting officer data cards shall be required as specified in MIL–A–2550.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility or any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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